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FACULTY OF COMPUTING
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SCSM2113 - HUMAN-COMPUTER INTERACTION

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PROJECT PROPOSAL

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Section: 01

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INTRODUCTION

The Digital Twin System has transformed the current industrial operations by offering virtual models of physical assets that mimic and predict the real-time performance. Although sectors such as manufacturing and smart cities have broadly embraced Digital Twin Systems, oil rigs have also started to incorporate these technologies to enhance in operations and safety.

However, the potency of the systems in the oil and gas sector is significantly influenced by the Human-Computer Interaction components of the system. Employees like the maintenance engineers, operators, and supervisors, regularly encounter difficulties when they are engaging with intricate data visualizations, coping with information overload, and sustaining confidence in automated results. This proposal addresses the Human-Computer Interaction challenges and proposes solutions specifically designed for oil rig environments.

PROBLEM STATEMENT

The integration of Digital Twin platforms, as proposed by DreamEDGE, offers transformative potential for the smart industry. However, the human-centred challenges of interacting with such data-rich systems are a critical barrier to adoption, especially in oil rigs. Our analysis, guided by the HCI challenges of Usability, Interaction, and Accessibility, has identified the following three core problems:

Problems	**Why is it a Problem?**	**Classification**	**Sustainable Development Goal (SDG)**
1.Core problem for mobile users. (Interaction Challenge) Maintenance engineers, trainees, and supervisors on offshore oil rigs work in physically demanding, high-risk environments. Current tools like desktop dashboards or handheld tablets create major bottlenecks.	• Workers must stop tasks, remove gloves, and divert attention to access schematics or diagnostics • Interruptions break workflow, reduce efficiency, and increase safety risks. • Supervisors struggle to provide real-time guidance, slowing decision-making and error correction.	User experience	This problem relates to **SDG 9: Industry, Innovation, and Infrastructure**, as solving this interaction bottleneck improves the safety and efficiency of industrial processes.
2.Overwhelming Dashboards with Too Much Data (Usability Challenge) In oil rig environments, digital twin dashboards often present large amounts of real-time operational data all at once—such as pressure readings, pump load metrics, vibration frequencies, temperature	• Supervisors may overlook critical safety alerts because they are hidden within layers of dense visual information. • Trainees and new operators struggle to understand what data is relevant, leading to confusion during fast-paced rig operations. • Managers who need quick	Usability	It relates to **SDG 9: Industry, Innovation, and Infrastructure**. Digital twin systems are part of advanced industrial infrastructure. But if non-technical users like supervisors, trainees, and managers cannot understand or act on the data, the innovation fails to be inclusive or effective.

fluctuations, and predictive equipment failure indicators. These dashboards become heavily saturated with charts, graphs, alarms, and technical overlays.	situational updates get slowed down by excessive technical details, affecting timely decision-making during high-risk scenarios.		
3.Lack of Accessibility and Inclusivity for Diverse User Groups Oil rigs employ a diverse workforce that includes technicians, operators, offshore engineers, contractors, and personnel with varying degrees of technical proficiency. Digital twin systems are often not designed to accommodate users with physical limitations, lower digital literacy, or multilingual needs, resulting in uneven access to system functions and insights.	The lack of accessibility and inclusivity in the digital twin interface means that many rig workers cannot fully utilize the system. This leads to operational inefficiencies, increased training burdens, misinterpretation of safety-critical information, and the exclusion of workers who require more supportive or adaptive interfaces to perform effectively.	User experience	This problem aligns with **SDG 9: Industry, Innovation, and Infrastructure**, as inclusive design is important for the successful adoption and efficiency of industrial digital systems.

PROPOSED SOLUTION

Our group is addressing the identified challenges by proposing a set of targeted solutions. Each solution is designed to solve a specific problem, supporting better interaction, comprehension, and decision-making for that context. The solutions are as follows:

Problem-solving 1: (Interaction Challenge):

Proposed Solution	**Description of the Solution**	**Enhancement Towards the Users**
"Context-Aware" AR Smart Helmet.	The proposed solution is an industrial-grade safety helmet equipped with a transparent augmented-reality (AR) visor. This AR visor seamlessly overlays essential digital twin information—such as real-time temperature readings, pressure data, equipment schematics, and step-by-step maintenance instructions—directly into the engineer's line of sight on the oil rig. This allows workers to access critical information without shifting attention away from the physical equipment in front of them.	The AR helmet keeps the engineer's hands free, using a multi-modal interaction system designed for harsh offshore environments. • Voice commands allow workers to request data, perform checks, or open instructions without interrupting their tasks. • A glove-friendly tactile side panel acts as a reliable backup in noisy rig conditions where voice input may fail. By delivering data at the exact point of work, the solution eliminates workflow interruptions, reduces cognitive load, and enhances accuracy and safety during critical maintenance or inspection activities.

Image Reference:

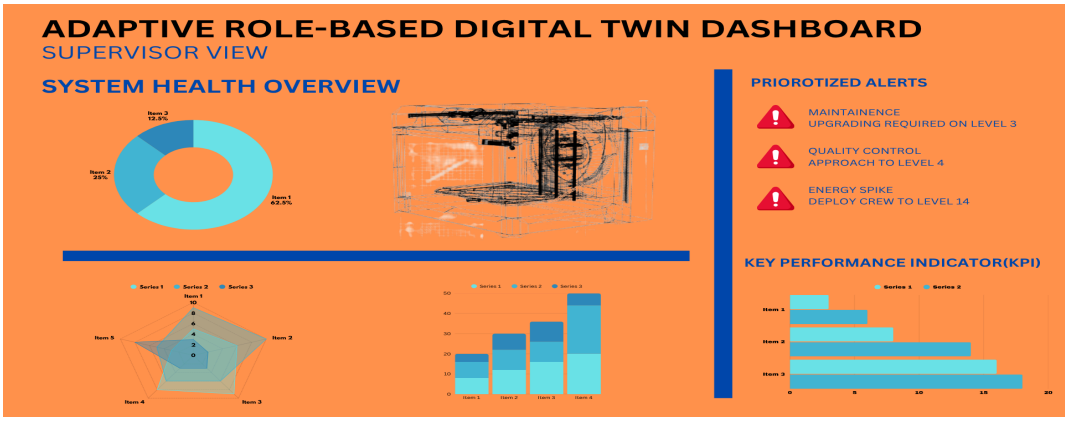


Problem-solving 2: (Usability Challenge):

Proposed Solution	Description of the Solution	Enhancement Towards the Users
A. Role-Based Personalization	The system provides customized interfaces tailored to the user's role within the oil rig operation.	• Supervisors: Receive high-level summaries, health overviews, and prioritized safety alerts for rapid decision-making. • Engineers: Access detailed diagnostics, trend visualizations, vibration analyses, and technical parameters for effective troubleshooting. • Trainees: View simplified dashboards with guided tooltips that clarify the meaning of each metric and alert. This ensures each user only sees what is relevant to their operational responsibility, helping reduce information overload and improving usability for all levels. •
B. Progressive Disclosure:	The dashboard uses a tiered information structure, starting	Users begin with a clean overview and then drill down into deeper data, such as historical trends or sensor-level diagnostics, only when necessary.

	with essential overview data and revealing increasing detail only when users choose to expand specific panels.	This preserves a clean interface, supports natural workflow exploration, and prevents the overwhelming presentation of data.
C. Visual Clarity and Information Hierarchy	The interface is redesigned using minimalist visual principles, with limited color palettes (e.g., green for normal conditions, red for critical issues), consistent iconography, and clearly grouped data clusters.	This approach improves visual clarity, allowing critical alerts to stand out immediately and strengthening situational awareness across supervisors, engineers, and trainees. Users can quickly interpret system conditions without being distracted by unnecessary visual noise.

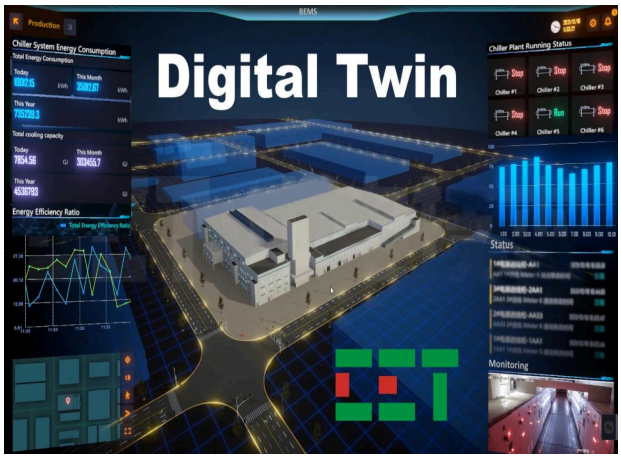
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Problem-solving 3: (Accessibility & Inclusivity Challenge):

Proposed Solution	Description of the Solution	Enhancement Towards the Users
A. Customizable Interfaces	The dashboard offers adjustable interface settings such as scalable font sizes, selectable color schemes, simplified layouts, and accessibility modes for users with specific visual or cognitive needs.	Users can personalize the interface according to their abilities—for example: ● High-contrast mode for users with dyslexia ● Colorblind-friendly modes ● Simplified tile layouts for workers with limited digital literacy These adjustments enhance navigation, reduce strain, and ensure that all workers can access and interpret information effectively.
B. Context-Sensitive Help and Tutorials	The system provides task-aware help guides in multiple formats including text, audio, and short instructional videos. Tutorials adapt to the user's current activity in the digital twin interface.	● This feature supports both novice and experienced users by offering real-time guidance appropriate to their workflow. It accommodates different learning styles, increases confidence, and helps ensure that no user—regardless of background or ability—is excluded from effectively interacting with the system.

Image Reference :



TARGET USERS

1. Maintenance Engineers

Maintenance engineers on the oil rig benefit directly from the **Context-Aware AR Smart Helmet**, which provides a hands-free, multi-modal interaction system. They can use **voice commands** to call up schematics, run diagnostics, or view live sensor data without pausing their work.

Additionally, through **Role-Based Personalization**, engineers have access to deeper diagnostic layers, vibration analyses, trend charts, and technical metrics—supporting more accurate and efficient troubleshooting in high-pressure environments.

2. Rig Supervisors

Rig supervisors gain value from both **Role-Based Personalization** and enhanced **Visual Clarity** in the redesigned dashboard. Their custom interface highlights only essential information such as equipment health summaries, operational KPIs, and prioritized safety alerts.

Using simplified color cues and organized data grouping, supervisors can instantly identify abnormal conditions, enabling faster, more informed decisions during routine monitoring or critical events.

3. Trainees & Users with Disabilities

Trainees and accessibility-focused users benefit from **Customizable Interfaces** and **Context-Sensitive Help** features.Trainees interact with simplified dashboards supported by guided tooltips and task-based tutorials that help them understand system behaviors without being overwhelmed.Users with disabilities—including those with dyslexia, color vision deficiencies, or visual impairments—can personalize the interface with adjustable font sizes, alternative color schemes, or high-contrast modes. These adaptations enhance readability, reduce cognitive strain, and ensure inclusive access to digital twin information across all worker groups.

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